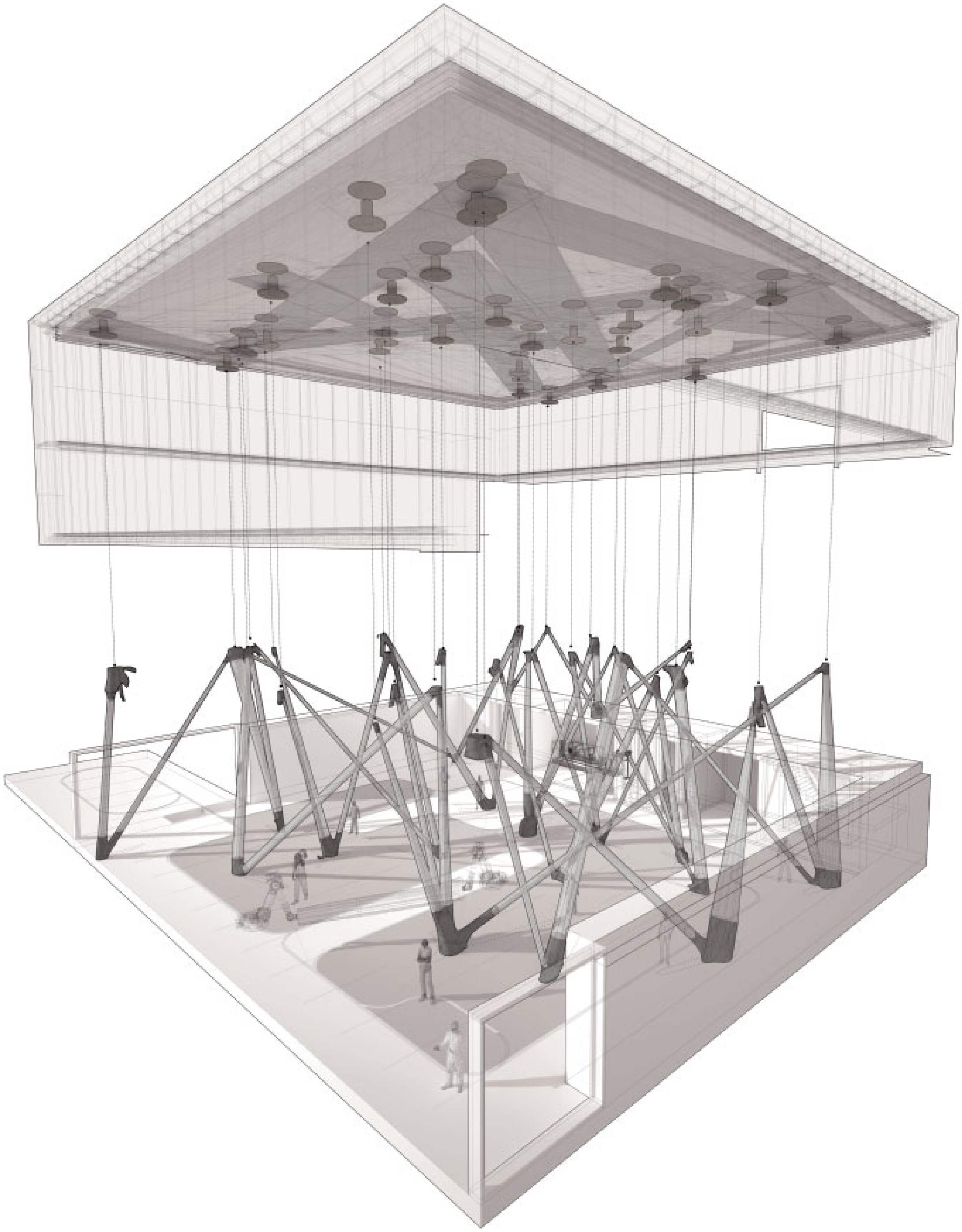
INTELLIGENT WOOD ASSEMBLIES: INCORPORATING FOUND GEOMETRY & NATURAL MATERIAL COMPLEXITY

The onset of digital technology has seen the embracing on non-standard design techniques that use standardised materials, typified by the use of milled plywood. Jonathan Enns describes a project that he has developed at Princeton seeking to harness the irregular and materially complex nature of the innate geometry in timber in such as way as to directly characterise the final design.

Jonathan Enns, Digital Fabrication Workshop, Thesis Design Proposal, Princeton University School of Architecture, Princeton, New Jersey, 2010
Exploded building perspective. The fabrication facility makes use of the anisotropic characteristics of timber, in both the linear elements and the joint transitions, to determine the arrangement of the structural frame. This acts as both support for the enclosure, and as a programmatic determinant.



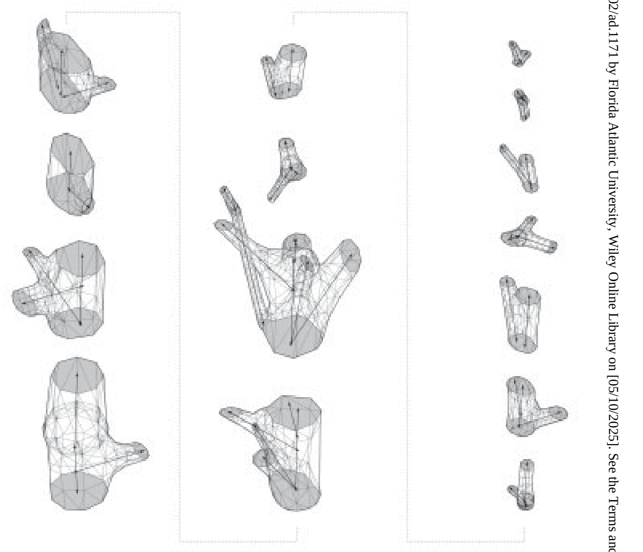
‘Nonstandard’ techniques have so far been celebrated as liberating alternatives to the homogenising formal tendencies of mass production. Nonstandard production, however, has continued to require the use of standardised input material, the prototypical example being the ubiquitous stacked plywood milled model. Such practices constitute an unnecessarily inefficient production circuit that moves from nonstandard input such as a tree to standardised stock material such as plywood or dimensional lumber, and back to nonstandard forms through digital fabrication.

A more direct translation was proposed in the design of a workshop building for digital fabrication, a project developed in the thesis programme at Princeton University School of Architecture in the spring of 2010.

The goal of the project was to test an alternative process for the creation of nonstandard form in which the irregular and materially complex nature of the found geometry would directly determine the final design. Unlike the standardising tendencies of current production processes, the proposed transfer, from stock material to finished design, attempted to minimise the loss of embodied material intelligence. Efforts were focused on retaining the anisotropic capabilities of timber, particularly in locations such as branch transitions, features that typically go unused in conventional design and production and constitute a major net loss in terms of material intelligence.

Precedents for the project exist in early ship construction, in which trees with forms closest to those required were selected. This constructive logic allowed flexibility in the final design, while accepting the idiosyncrasies of the selected material. The ambition in the design of the workshop building was to adopt and retune similar logics to the capabilities of digital design and manufacturing. Such a process attempts to offer alternatives to current practices both in terms of material efficiency and in opening new potentials for formal exploration.

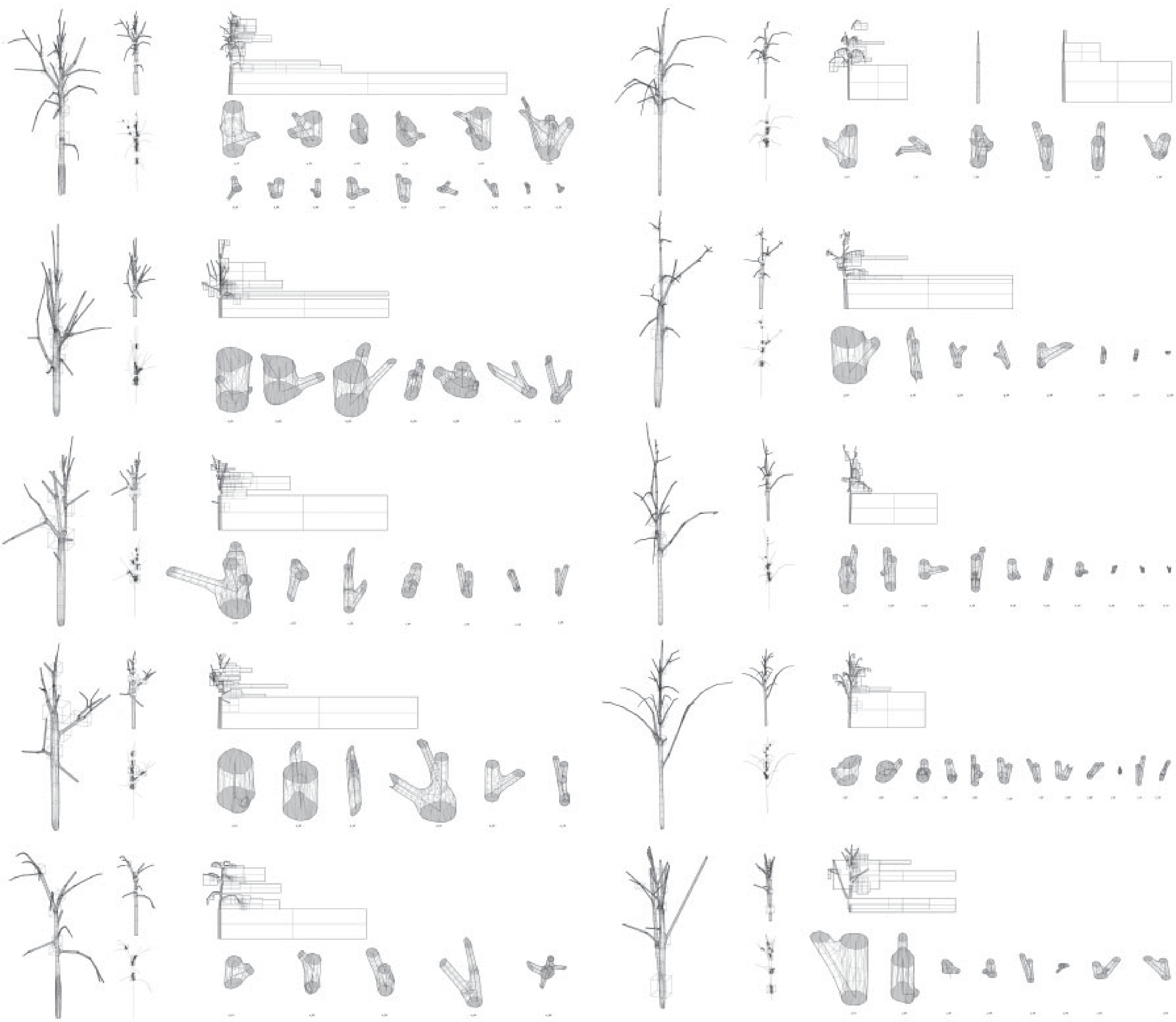
The workshop building houses an assembly floor for the production of digitally fabricated furniture on a heavily wooded site. At the outset, 10 trees occupying the building footprint were selected to be documented from multiple viewpoints with controlled-perspective photography. The images were then used

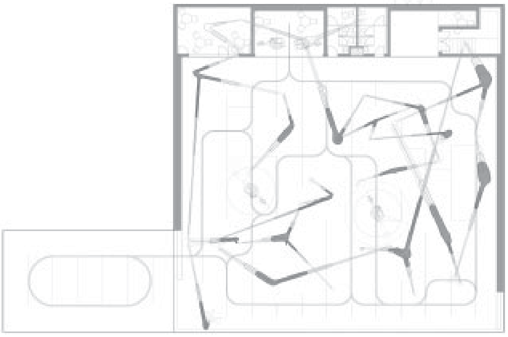


opposite top: Joint stock and sheets stock for tree number 1. The digital instantiation of the stock trees allows the dimensions, quantities and types of available materials to be understood and manipulated before cuts to the source tree are made. Here, the types of material consist of joint stock and sheet stock, the latter produced through radial cutting. This is visualised as the potential dimension of the 'unrolled' timber (bottom drawing).

opposite bottom: Visualisation of site-found timber stock (tree number 1 of 10) and locations of sheet and joint elements. Locations for sheet stock utilise the linear elements of the tree between branch nodes, while the joint stock makes use of the intelligent anisotropic grain distributions in the joints. The latter is not typically used in timber manufacturing processes.

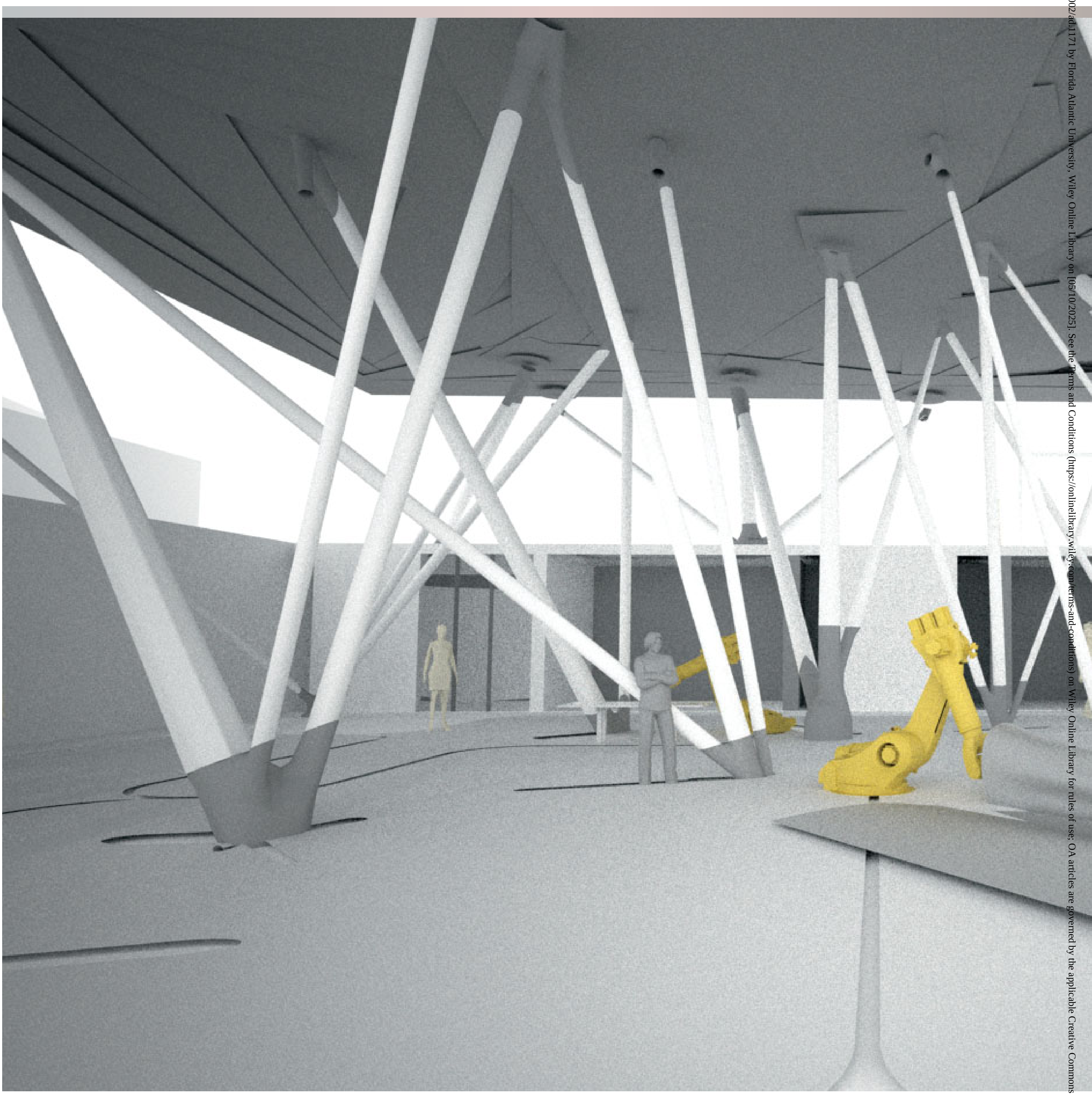
below: Digital tree inventory. Living site trees were instantiated into a digital catalogue where joint and sheet stock quantities could be calculated, visualised and manipulated in the design. This process allows the initial cuts to the tree to be custom-tailored to the final material arrangement.

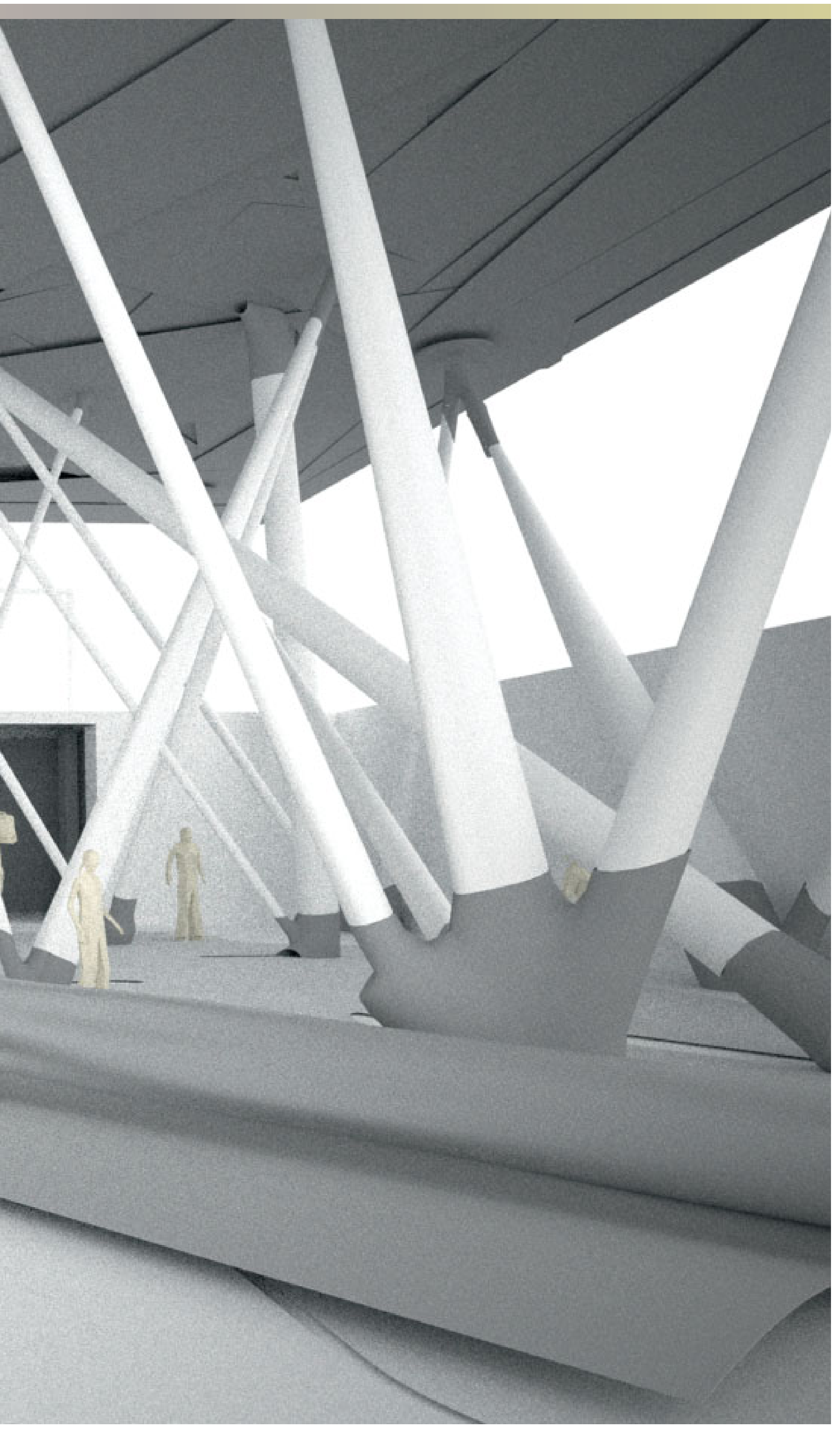




left: Building plan. The workshop functions as a large assembly floor for digitally fabricated furniture. The individual geometries of the tree joints influence the arrangement of the structural frame, which in turn determines the layout of the plan through head clearance and allowances for assembly tracks.

below: Interior rendering of the workshop building. Sheet stock is distributed in the roof in a 'tuned-ply' arrangement, and in the body of the columns (rendered in white). The joints are located at the top and bottom of the structural frame (rendered in darker grey). The joints' individual geometries determine the trajectories and overall arrangement of the structural frame.





for digital reconstructions of the site trees and instantiation into a 'digital catalogue'. From these base instantiations, both linear sections and branching nodes were located and separated as independent lists of geometry. These catalogues were operated upon using scripting and parametric software, providing quantitative readouts such as dimension, angle, volume and so on. The cataloguing of information enabled the manipulation of the irregular geometry and design progression before any custom cuts to the living trees were made.

While the joint-stock catalogue provided lists of angles and cross-sectional areas for each joint on each tree – determining the trajectories and spans in the structural frame – the linear-stock catalogue provided information on the potential sheet material resulting from radially cutting these latter elements. This information allowed the calculation of timber volume, carbon storage and total surface area coverage, as well as dimensions for the arrangement of lamination in the roof and column bodies. Elements from the joint-stock catalogue were located at the top and bottom terminations of the frame, with each exiting trajectory aligned to the entry trajectory of the proceeding joint, and so on. These were connected with laminated tubes of sheet-stock material with longitudinal grain alignment, producing an entirely anisotropic structural frame.

The unmediated use of nonstandard input material in the creation of nonstandard form holds potential for material efficiency and formal exploration, but also has further ideological implications. By resisting the totalising control of the author through the contribution of found geometries, such a project places emphasis not on imposed formal concepts, but rather on the design, capability and robustness within the mechanisms of control. This emphasis trades idealised regularising concepts such as repetition, sameness or exactitude – qualities that inherently resist the inclusion of found geometry – for pliant ones such as association, similarity and approximation in a system that provides for a type of reliable outcome but always a unique and never a certain one. ⅅ